Section 1.4 Logarithms & exponentials

对数与指数

Method 1 / 方法 1 [AL04-M01]

Take domains seriously, then let logarithms turn multiplication into addition or exponentials turn a shared exponent into one positive variable. Combine identities before solving, and distinguish existence from uniqueness.

先认真处理定义域，再用对数把乘法变为加法，或用指数把共同指数化为一个正变量。求解前先合并恒等式，并区分存在性与唯一性。

Change base and combine powers before solving variables

求变量前先换底并合并幂

Procedure

1. Use $\log_a b = \ln b / \ln a$ with $a > 0, a \neq 1, b > 0$.

2. Use reciprocal logarithms and $c^{\log_c d} = d$ to simplify the target directly.

3. Combine weighted log sums as a single product logarithm; retain positivity when extracting real powers.

通用步骤

1. 在 $a > 0, a \neq 1, b > 0$ 下使用 $\log_a b = \ln b / \ln a$。

2. 使用互反对数及 $c^{\log_c d} = d$，直接化简目标。

3. 把加权对数和合成乘积的对数；取实幂时保留正性。

> Example. Evaluate $(\log_2 9)(\log_3 4)$.

> 例题：计算 $(\log_2 9)(\log_3 4)$。

Section 1.4 Logarithms & exponentials

对数与指数

Method 1 / 方法 1 [AL04-M01] (continued / 续)

Worked solution

1. Extract powers: $\log_2 9 = 2\log_2 3$ and $\log_3 4 = 2\log_3 2$.

2. Change of base gives $(\log_2 3)(\log_3 2) = 1$.

3. The result is 4. All logarithm bases and arguments are valid.

例题解析

1. 提出幂：$\log_2 9 = 2\log_2 3, \log_3 4 = 2\log_3 2$。

2. 换底得 $(\log_2 3)(\log_3 2) = 1$。

3. 结果为 4，各底数与真数均有效。

> A logarithm of a sum does not split into a sum of logarithms.

> 和的对数不能拆为对数之和。

Practice / 配套练习: AIME-2000-II-Q01 (WB p. 99), AIME-2003-I-Q04 (WB p. 100), AIME-2006-II-Q02 (WB p. 102), AIME-2009-II-Q02 (WB p. 103), AIME-2015-II-Q04 (WB p. 262), AIME-2024-I-Q02 (WB p. 113), AIME-2002-II-Q03 (WB p. 529)

Section 1.4 Logarithms & exponentials

对数与指数

Method 2 / 方法 2 [AL04-M02]

Use logarithmic coordinates and a shared value

使用对数坐标与共同值

Procedure

1. Choose one fixed base and replace positive variables by their logarithms.

2. Products and quotients become linear sums and differences; a common variable-base logarithm can be denoted by one shared value.

3. Add or subtract the resulting equations to eliminate repeated coordinates or a repeated exponential term.

4. Reconstruct positive original variables and verify every original base differs from one.

通用步骤

1. 选择统一底数，把正变量换为其对数。

2. 乘除变为线性和差；变底对数的共同值可统一设为一个变量。

3. 将所得方程相加或相减，消去重复坐标或共同指数项。

4. 还原正的原变量，并检查每个原底数不为一。

> Example. Positive x, y satisfy $\log_2(xy) = 5$ and $\log_2(x/y) = 1$. Find x, y.

> 例题：正数 x, y 满足 $\log_2(xy) = 5, \log_2(x/y) = 1$，求 x, y。

Worked solution

1. Put $u = \log_2 x, v = \log_2 y$. The equations become $u + v = 5, u - v = 1$.

Section 1.4 Logarithms & exponentials

对数与指数

Method 2 / 方法 2 [AL04-M02] (continued / 续)

2. Adding and subtracting gives $u = 3, v = 2$.

3. Hence $x = 8, y = 4$; their product and quotient are $32, 2$ as required.

例题解析

1. 令 $u = \log_2 x, v = \log_2 y$，方程化为 $u + v = 5, u - v = 1$。

2. 相加与相减得 $u = 3, v = 2$。

3. 故 $x = 8, y = 4$，积与商分别为 $32, 2$，符合条件。

The logarithmic variables may be negative even though the original variables are positive.

原变量为正，对数变量仍可能为负。

Practice / 配套练习: AIME-2000-I-Q09 (WB p. 116), AIME-2002-I-Q06 (WB p. 117), AIME-2006-I-Q09 (WB p. 474), AIME-2010-II-Q05 (WB p. 105), AIME-2012-I-Q09 (WB p. 120), AIME-2016-II-Q03 (WB p. 107), AIME-2019-I-Q07 (WB p. 477), AIME-2019-II-Q06 (WB p. 123), AIME-2022-II-Q04 (WB p. 111), AIME-2024-II-Q04 (WB p. 114)

Section 1.4 Logarithms & exponentials

对数与指数

Method 3 / 方法 3 [AL04-M03]

Turn an exponential equation into a polynomial on a restricted range

把指数方程转为受限范围内的多项式

Procedure

1. If the exponents share a nonconstant linear expression, put $t = b^{cx}$ with $b > 0, b \neq 1, c \neq 0$. For unrestricted real x its exact range is $t > 0$; intersect this with any original restrictions on x.

2. Solve or analyze the resulting polynomial; discard nonpositive roots before returning to x.

3. Use Vieta for a sum of logarithms or product of original variables, avoiding individual roots when possible.

4. For equations such as $x^y = y^x$, take logarithms or a positive real root only within the original positive domain.

通用步骤

1. 若指数共享非恒定线性表达式，可令 $t = b^{cx}$，其中 $b > 0, b \neq 1, c \neq 0$。当 x 遍历实数时其准确值域为 $t > 0$；若原 x 有限制，还须取相应交集。

2. 求解或分析所得多项式，返回 x 前剔除非正根。

3. 对数和或原变量乘积可用韦达，尽量避免逐个求根。

4. 对 $x^y = y^x$ 等式，只在原正性定义域内取对数或正实根。

> Example. Classify the number of real solutions to $4^x - a2^x + 1 = 0$ as real a varies.

> 例题：按实参数 a 分类 $4^x - a2^x + 1 = 0$ 的实数解个数。

Section 1.4 Logarithms & exponentials

对数与指数

Method 3 / 方法 3 [AL04-M03] (continued / 续)

Worked solution

1. Put $t = 2^x > 0$. The equation is $t^2 - at + 1 = 0$.

2. Real roots require $|a| \geq 2$. Their product is one and sum is a, so they are positive exactly when $a \geq 2$.

3. At $a = 2$ there is one root $t = 1$, giving one real x; at $a > 2$ there are two; for $a < 2$ there are none.

例题解析

1. 令 $t = 2^x > 0$，方程为 $t^2 - at + 1 = 0$。

2. 实根要求 $|a| \geq 2$；根积为一、根和为 a，所以恰在 $a \geq 2$ 时为正。

3. $a = 2$ 时只有根 $t = 1$，对应一个实数 x；$a > 2$ 时有两个；$a < 2$ 时没有。

A real polynomial root may lie outside the range of the exponential substitution.

多项式的实根可能位于指数代换的值域之外。

Practice / 配套练习: AIME-2005-I-Q08 (WB p. 118), AIME-2005-II-Q05 (WB p. 101), AIME-2010-I-Q03 (WB p. 104), AIME-2018-I-Q05 (WB p. 108), AIME-2020-I-Q02 (WB p. 109), AIME-2023-I-Q02 (WB p. 112), AIME-2026-I-Q06 (WB p. 124)

Section 1.4 Logarithms & exponentials

对数与指数

Method 4 / 方法 4 [AL04-M04]

Undo nested logarithms and prove uniqueness monotonically

逐层还原嵌套对数，并用单调性证明唯一

Procedure

1. Work from the outermost equation inward, recording positivity of every inner argument.

2. If the base is unknown, a reciprocal logarithm variable can turn the remaining relation into a scalar equation.

3. Compare monotonicity of the two sides to prove at most one admissible solution; then verify a candidate exactly.

4. For integer base exponents, reduce the result to prime-exponent or bounded divisor conditions.

通用步骤

1. 从最外层方程向内处理，记录每个内层真数的正性。

2. 底数未知时，可用互反对数变量把剩余关系变成单变量方程。

3. 比较两侧单调性，证明至多一个允许解，再精确验证候选。

4. 若底数指数要求为整数，将结果化为质指数或有界约数条件。

> Example. Find the base $b > 1$ for which $\log_b(\log_b 16) = 2$.

> 例题：求使 $\log_b(\log_b 16) = 2$ 的底数 $b > 1$。

Section 1.4 Logarithms & exponentials

对数与指数

Method 4 / 方法 4 [AL04-M04] (continued / 续)

Worked solution

1. Exponentiating the outer logarithm gives $\log_b 16 = b^2$, or $\ln 16 = b^2 \ln b$.

2. For $b > 1$, both positive factors $b^2, \ln b$ strictly increase, so their product strictly increases.

3. The value $b = 2$ works because $4 \ln 2 = \ln 16$. Monotonicity proves it is unique, and the inner logarithm equals four and is positive.

例题解析

1. 还原外层对数得 $\log_b 16 = b^2$，即 $\ln 16 = b^2 \ln b$。

2. 当 $b > 1$，正因子 $b^2, \ln b$ 均严格增加，所以乘积严格增加。

3. $b = 2$ 成立，因为 $4 \ln 2 = \ln 16$；单调性证明唯一，内层对数为正的四。

Checking one guessed base establishes existence; monotonicity establishes uniqueness.

验证猜到的底数只证明存在，单调性才证明唯一。

Practice / 配套练习: AIME-2013-II-Q02 (WB p. 106), AIME-2016-II-Q03 (WB p. 107), AIME-2017-I-Q14 (WB p. 125)

Section 1.4 Logarithms & exponentials

对数与指数

Method 5 / 方法 5 [AL04-M05]

Partition logarithmic floors at exact power thresholds

按精确幂门槛分割对数取整

Procedure

1. Use $\lfloor \log_b t \rfloor = k \iff b^k \le t < b^{k+1}$ for $b > 1$.

2. For a floor in an exponent, partition the original variable into its unit intervals before evaluating its image.

3. Convert the problem into interval lengths, integer counts or a monotone threshold sum; preserve half-open endpoints.

4. At a proposed greatest integer, verify the neighboring integer and invoke monotonicity to exclude all later ones.

通用步骤

1. 当 $b > 1$，使用 $\lfloor \log_b t \rfloor = k \iff b^k \le t < b^{k+1}$。

2. 指数中含取整时，先将原变量按单位区间分块，再求其像。

3. 将题目化为区间长度、整数计数或单调门槛和，并保留半开端点。

4. 拟定最大整数后，检验相邻整数，再用单调性排除其后全部整数。

> Example. Choose x uniformly from $[1, 4]$. Find the probability that $\lfloor \log_2(3x) \rfloor - \lfloor \log_2 x \rfloor = 1$.

> 例题：从 $[1, 4]$ 按长度均匀取 x，求 $\lfloor \log_2(3x) \rfloor - \lfloor \log_2 x \rfloor = 1$ 的概率。

Section 1.4 Logarithms & exponentials

对数与指数

Method 5 / 方法 5 [AL04-M05] (continued / 续)

Worked solution

1. On $[1,2)$ the second floor is zero. The first floor is one precisely on $[1,4/3)$.

2. On $[2,4)$ the second floor is one. The first floor is two precisely on $[2,8/3)$.

3. The favorable lengths total $1/3+2/3=1$. Divide by sample length three to obtain $1/3$; endpoints have zero measure.

例题解析

1. 在 $[1,2)$ 上，第二个取整为零；第一个恰在 $[1,4/3)$ 为一。

2. 在 $[2,4)$ 上，第二个取整为一；第一个恰在 $[2,8/3)$ 为二。

3. 有利长度合计 $1/3+2/3=1$，除以样本长度三得 $1/3$，端点测度为零。

A ceiling-floor difference vanishes at integral logarithms; these are exact powers, not approximate ones.

上取整减下取整在整数对数处为零；这些是精确幂，不能以近似判断。

Practice / 配套练习: AIME-2004-I-Q12 (WB p. 171), AIME-2007-I-Q07 (WB p. 163), AIME-2009-I-Q06 (WB p. 166), AIME-2010-I-Q14 (WB p. 175)

Section 1.4 Logarithms & exponentials

对数与指数

Method 6 / 方法 6 [AL04-M06]

Telescope logarithmic products and exponent progressions

使对数乘积与指数等差式望远镜消去

Procedure

1. Change all logarithms to one base and factor rational exponents before multiplying.

2. Cancel consecutive ratios over the full index range, retaining the first and last factors.

3. For a geometric sequence of powers, its logarithms form an arithmetic sequence; compare a positive sum with its largest term using a rigorous geometric tail bound.

通用步骤

1. 所有对数换至统一底数，并在相乘前分解有理指数。

2. 在完整下标范围内消去相邻比值，保留首尾因子。

3. 幂的等比数列取对数后为等差数列；正项和与最大项之间用严格等比尾和界比较。

> Example. Evaluate $\prod_{k=2}^{7} \log_k(k+1)$.

> 例题：计算 $\prod_{k=2}^{7} \log_k(k+1)$。

Worked solution

1. The kth factor is $\ln(k+1)/\ln k$.

2. All intermediate logarithms cancel, leaving $\ln 8/\ln 2$.

3. The product equals 3. Every base in the range is positive and different from one.

Section 1.4 Logarithms & exponentials

对数与指数

Method 6 / 方法 6 [AL04-M06] (continued / 续)

例题解析

1. 第 k 个因子为 $\ln(k+1)/\ln k$。

2. 所有中间对数相消，留下 $\ln 8/\ln 2$。

3. 乘积等于 3；范围内所有底数均为正且不为一。

Cancel factors across products, never terms across sums.

只能在乘积中约去因子，不能在和式中随意消项。

Practice / 配套练习: AIME-2007-II-Q12 (WB p. 159), AIME-2009-I-Q07 (WB p. 147), AIME-2014-II-Q07 (WB p. 149), AIME-2025-II-Q04 (WB p. 115)

Section 1.4 Logarithms & exponentials

对数与指数

Method 7 / 方法 7 [AL04-M07]

Enforce logarithm domains before inequalities and root counts

先施加对数定义域，再处理不等式与根数

Procedure

1. List positive arguments, positive bases different from one, and any inverse-trigonometric range restrictions.

2. When exponentiating inequalities, distinguish bases above one from bases between zero and one.

3. After a substitution or squaring, test every branch in the original logarithmic domain.

4. Only then count integer parameters or admissible real roots.

通用步骤

1. 列出真数为正、底数为正且不为一，以及反三角函数的范围限制。

2. 指数化不等式时，区分大于一的底数与零一之间的底数。

3. 代换或平方之后，将每个分支代回原对数定义域检验。

4. 之后才统计整数参数或允许实根。

> Example. Find the real domain of $\arcsin(\log_{1/2} x)$.

> 例题：求 $\arcsin(\log_{1/2} x)$ 的实数定义域。

Worked solution

1. The logarithm requires $x > 0$ and its fixed base is valid.

Section 1.4 Logarithms & exponentials

对数与指数

Method 7 / 方法 7 [AL04-M07] (continued / 续)

2. The arcsine requires $-1 \le \log_{1/2} x \le 1$. Since the base is below one, exponentiation reverses the order.

3. Thus $1/2 \le x \le 2$, with both endpoints allowed.

例题解析

1. 对数要求 $x > 0$，固定底数有效。

2. 反正弦要求 $-1 \le \log_{1/2} x \le 1$；底数小于一，指数化会反转顺序。

3. 因此 $1/2 \le x \le 2$，两端点均允许。

The algebraic equation alone cannot certify that its logarithms exist.

单有代数方程不能保证其中的对数存在。

Practice / 配套练习: AIME-2005-II-Q05 (WB p. 101), AIME-2009-II-Q11 (WB p. 174), AIME-2011-I-Q09 (WB p. 119), AIME-2013-I-Q08 (WB p. 121), AIME-2017-II-Q07 (WB p. 122), AIME-2018-I-Q05 (WB p. 108), AIME-2020-I-Q02 (WB p. 109), AIME-2020-II-Q03 (WB p. 110), AIME-2023-I-Q02 (WB p. 112)